Project Proposal

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1 Title and Team Information

Title: Using Regression Models to Predict Music Popularity

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2 Research Question and Motivation

2.1 Research Question

Can we use machine learning to accurately predict whether a song will be popular?

2.2 Motivation

Music is everywhere in modern life, and yet most people only hear a small percentage of all music created. There must be a reason for some music being wildly more popular than others, and our project intends to shed some light on why. Also, music producers may want to predict if a new song could potentially be a hit, and our classifier may be able to predict it before the song is released.

3 Data Sources

3.1 Data Source

The Spotify Tracks Dataset will serve as our dataset for this project. The dataset can be found on Kaggle and downloaded as a CSV. The main columns we plan on using from the dataset are popularity, danceability, energy, loudness, tempo, valence, instrumentalness, and speechiness. The data includes songs from multiple languages, and it was last updated in 2022.

3.2 Database Download Link

`https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset/data`

3.3 Preprocessing

We will use the Pandas library through Python to import the data as a dataframe. Cleaning and preprocessing will be a simple process of using Pandas functions to remove null values, duplicates, and irrelevant columns from the data. The popularity column will be our main target variable for classification. The Spotify Tracks Dataset scores popularity on a range from 0 to 100.

38th Conference on Neural Information Processing Systems (NeurIPS 2024).

We can choose ranges to decide what is popular and what is not popular. For our purposes, a range from 0 to 69 will be deemed unpopular, while 70 and above will be popular. Since the features are ordered on different scales (0-1 for danceability, 0-100 for popularity, etc), we will have to standarize the data. We will accomplish this by using Scikit-Learn's StandardScaler to transform features so they have a mean of zero and a standard deviation of one. This ensures no one feature influences the data more than others.

4 Methodology and Analysis Plan

4.1 Methodology

The classification task will be approached using the following machine learning models: Logistic Regression will be used as a baseline model due to its interpretability and effectiveness with numerical data. In addition, a Random Forest classifier will be implemented to capture nonlinear relationships between audio features and song popularity. These models will be trained using Spotify audio features such as danceability, energy, tempo, loudness, and valence.

4.2 Analysis Plan

Model performance will be evaluated using metrics including accuracy, precision, recall, and F1 score.

5 Expected Outcomes and Evaluation

5.1 Expected Outcomes and Evaluation

A songs popularity has many factors, so it will be interesting to find out what aspects of a song's composition push it towards popularity, and which ones hinder it. We believe that we will find that a song's danceability, energy, and tempo will be the biggest contributors to a song's potential popularity. More specifically, the higher these features are, the more likely that song is to be popular. To assess our results, we will look at how well our model is able to predict a hit when compared with the ground truth. Metrics such as accuracy, precision, recall, and F1 score will tell us how well our model classifies a song.

5.2 Extensions

One possible extension could be to include more data on other aspects of a songs popularity, outside of what our data set provides, like artist's popularity, genre, or advertising budget. Another extension could be to incorporate a CNN that processes the audio spectrogram into the model to allow for a better analysis of each song. A Many-to-One RNN could also be used to define lyrics sentiment for each song. This way, we could find out what topics of songs are more popular. In addition, our data set ends at the year 2022, so incorporating newer data may improve model performance and give it more ability to predict modern trends.